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# From Hand-Drawn Schematics to Structural Circuit Netlists: A Deterministic Pipeline with Endpoint-Graph Wire Joining and a Human-Verified Connectivity Benchmark

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**ABSTRACT** We present a deterministic pipeline for structural circuit-netlist recovery using an occlusion-first wire extractor, an endpoint graph, and degree-budget completion. The output represents inferred pin-to-node assignments; our component-pair metric does not certify pin identities, exact net partitions, absence of shorts, or simulation equivalence. Illustrative SPICE export requires externally specified values and device models; simulation equivalence on real scans is not validated. We provide a synthetic perturbation protocol and a human-verified connectivity benchmark of 31 CGHD-1152 images. With annotated component boxes and detected wires, the join achieves component-pair micro-F1 0.890 (macro-F1 0.901), compared with 0.805 for Hough plus proximity and 0.624 for connected-component tracing on the same detected wires. The wire extractor achieves F1 0.976 on 134 images with annotated component boxes under the historical benchmark configuration. Under the same component priors on the 31-image benchmark, a vision-language model reaches micro-F1 0.923; the paired difference is +0.033, with 95% confidence interval  $[-0.009, +0.078]$ . Inclusion of zero does not establish equivalence or comparative cost superiority. These results provide within-corpus evidence, not validation of autonomous reconstruction or unseen-style generalization.

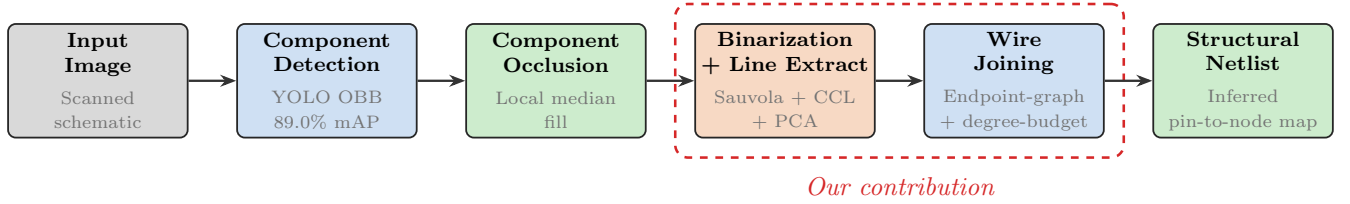
**INDEX TERMS** circuit digitization, connectivity benchmark, graph-based connectivity, hand-drawn circuits, image-to-netlist, netlist extraction, schematic recognition, SPICE, wire extraction, wire joining

## I. INTRODUCTION

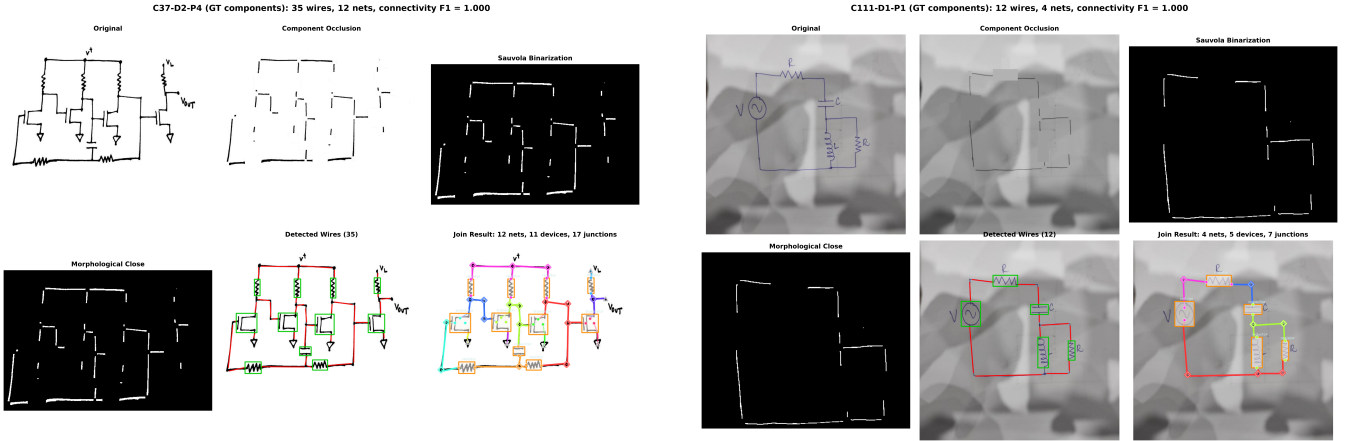
Analog circuit schematics encode component types, values, and topological connections in a visual format designed for human interpretation. Recovering a structural netlist from a scan is a prerequisite for simulation, design verification, and large-scale circuit analysis, but a complete simulatable SPICE (Simulation Program with Integrated Circuit Emphasis) deck also requires component values and device models. This work addresses the structural connectivity stage only. The *joining* problem determines which wire segments connect to which component pins and to each other. We study this stage with annotated component boxes; its relative contribution to autonomous reconstruction error is not established by that experiment.

The challenge is threefold. First, wire endpoints extracted from binarized images are inherently noisy: the same circuit drawn by different drafters produces endpoint positions that vary by 10–80 pixels. Second, components occupy bounding boxes that overlap with wire endpoints, creating ambiguity about whether an endpoint connects to a component's pin or merely passes nearby. Third, T-junctions, collinear fragments, and crossing wires create connectivity ambiguities for proximity-based rules, including the graph rules studied here.

A complementary approach is to prompt a vision-language model (VLM) for connectivity. Section IV-H reports an oracle-component-box diagnostic in which both methods receive the same annotated component boxes and the geometric method uses detected wires. The observed component-pair



**FIGURE 1.** Pipeline for structural circuit-netlist recovery. The highlighted stages extract wire endpoints and infer connectivity. The net-level experiments use annotated component boxes to isolate joining; they do not establish an end-to-end error hierarchy or validate illustrative SPICE export.



**FIGURE 2.** Pipeline stages on two real hand-drawn circuit images, joined with our scale-relative completion join (connectivity  $F1 = 1.000$  on both). Left: a four-transistor analog cell (35 wires, 7 nets); right: an RLC loop photographed on a textured surface (11 wires, 4 nets). Each panel shows, in order: original scan, component occlusion, Sauvola binarization, morphological close, detected wires (red overlay), and the join result. In the join result every recovered net is a distinct colour drawn along its conductors; component bounding boxes are the ground-truth labels (as used throughout the net-level evaluation, isolating the join), filled dots are device pins, diamonds mark junctions, and triangles mark terminals and ground.

scores do not establish equivalence, comparative cost superiority or a global structural-validity guarantee. The diagnostic examines the methods under those stated input conditions.

We address these challenges with a deterministic structural-netlist pipeline whose proposed elements are: (1) an *occlusion-first wire extractor* that masks detected components before binarization and estimates per-segment endpoints, producing a line-segment-with-endpoints representation (rather than tracing nets directly into connected components, as prior systems do) that is what makes graph-based joining possible; (2) an *endpoint-graph join* model that constructs a graph over both wire endpoints and component pins, with five edge types capturing distinct connectivity patterns; and (3) a *degree-budget completion* algorithm that proposes recovery connections using bounded assignment and merge guards. We evaluate on the CGHD-1152 dataset [1] with a synthetic error injection framework and a human-verified net-level benchmark. We evaluate topological recovery with illustrative SPICE export; component values and device models require external specification, and simulation equivalence on real scans is not validated. The output records inferred pin-to-node assignments, whereas the primary metric measures component-pair connectivity rather than exact pin topology.

## II. RELATED WORK

The end-to-end conversion of schematic images into machine-readable netlists has progressed rapidly. Kelly and Cole [2] presented a pattern-recognition workflow for digitizing circuit schematics with an 86.4% success rate, combining optical character recognition (OCR) for component values with network-graph construction. SINA [3], the closest contemporary system, pairs a YOLOv11 detector with connected-component labeling (CCL) and OCR, and invokes a vision-language model (VLM) for reference-designator assignment, reporting 96.47% netlist-generation accuracy, a  $2.72\times$  improvement over the prior state of the art. SINA confines the VLM to symbolic labeling and derives nets from CCL proximity rather than trusting the VLM with connectivity, an insight we share (Section II). Masala-CHAI [4] (Auto-SPICE) uses large language models to produce SPICE netlists from analog schematics at scale. Most directly comparable to our setting, the concurrent work of Peker *et al.* [5] (IEEE Access, 2026) builds a fully automated image-to-SPICE pipeline for printed and hand-drawn transistor-level circuits, pairing YOLOv8/11 detection with a *contour-based* node-detection module (components are erased and nets are read from the contours of the residual wire pixels) and validates netlists by automated LTspice simulation against a reference. On their in-house seven-topology MOSFET dataset they

report 93.33% (printed) and 85.33% (hand-drawn) whole-netlist success. Their node detection is an instance of the connected-component/contour recipe we evaluate as a baseline (Section IV-D); because their dataset (transistor-level analog, authored in-house) and metric (LTspice node-voltage pass/fail) differ from ours (discrete components on CGHD-1152, component-pair F1), the headline numbers are not directly comparable. Our controlled comparison evaluates a CCL baseline on identical detected wires, rather than reproducing their complete system. Instance-segmentation approaches extract circuit graphs directly from pixels: Bayer *et al.* [8] segment components and wires for graph extraction over the CGHD lineage. For hand-drawn inputs, object-detection pipelines [9] and the recent JUHCCR-v1 database [10] target component recognition on freehand sketches. Most of these pipelines build on general-purpose object detectors such as the YOLO family [11], [12] and OCR engines such as Tesseract [13], with the ultimate goal of producing simulator-ready descriptions in formats such as SPICE [14]. These works address detection, recognition and connectivity under different datasets and metrics. Our contribution evaluates joining under specified component priors, without treating detection or recognition as solved. SPICE-level circuit modeling likewise underpins application deployment in neighboring hardware domains, including memristor-based neuromorphic circuits that realize brain-inspired learning in hardware [6] and memristive decision-making frameworks for autonomous navigation [7].

Progress depends on annotated corpora. The CGHD hand-drawing dataset [1] provides component- and wire-level annotations for handwritten schematics. In the analog/mixed-signal (AMS) domain, AMSnet 2.0 [15] contributes a large database built with AI-assisted segmentation for net detection, while AMSBench [16] is a benchmark for evaluating multimodal LLM capabilities across AMS circuit tasks. Beyond circuits, DiagramNet [17] introduces an end-to-end framework and dataset for non-standard system-level diagrams, including an explicit connection-recognition task scored by F1. These benchmarks provide context for component recognition and connectivity evaluation. Our synthetic perturbation protocol isolates joining under authored errors; it does not measure the relative autonomous error contributions of those stages.

General-purpose VLMs and LLMs have been applied broadly to electronic design [18], and to circuit understanding in particular. Kulkarni *et al.* [19] proposed Near Sight Correction using VLMs, reporting 87.4% accuracy on circuit visual question answering (VQA) and 0.87 F1 on connection-matrix prediction. Connectivity-specific benchmarks report results that depend on the task, model and input conditions. On the DiagramNet connection task [17], leading general models score very low F1 (GPT-5 at 0.029, Claude-Sonnet-4 at 0.265, and Gemini-2.5-Pro at 0.008), whereas a task-tuned 3B model reaches 0.735. AMSBench [16] reports a similar gap for MLLMs on AMS connectivity. These task-specific results motivate a separately scoped connectivity diagnostic;

they do not establish that geometry is superior to VLMs on our corpus. Our Claude-VLM experiment uses matched annotated component priors, as described in Section IV-H.

Classical wire extraction relies on edge detection, morphological operations, and the Hough transform [20], typically preceded by adaptive binarization such as Sauvola thresholding [21] and connected-component labeling [22]. Our legacy join baseline uses radius-based proximity: each wire endpoint grabs all component pins within a fixed threshold (typically 30px), connected via transitive union-find [23]. This can over-merge when multiple components fall within the threshold, assigning both pins of a two-terminal part (R, C, L, D) to the same net. SINA [3] likewise derives nets from CCL proximity, which is similarly vulnerable to over-merge under detector noise. We instead model joining as a typed endpoint graph that incorporates component geometry and directional preference, and resolve residual ambiguity with a degree-budget completion: a bounded *b*-matching formulation [24], [25] that caps per-pin degree and thereby limits over-merge when endpoints are dropped or displaced. This differs from graph neural networks for netlist reverse engineering [26], which learn representations over already-extracted gate-level graphs; our graph is a purely *geometric* model whose edges encode spatial relationships between wire endpoints and component pins. A complementary recent direction replaces rule-based connectivity with a *learned* model: Netlistify [27] predicts terminal connectivity with a Transformer and reports a 12.4% F1 gain over AMSnet on printed AMS schematics. Such models use task-specific connectivity training data. Transfer of the cited printed-schematic model to this hand-drawn benchmark is not established here; our geometric join requires no connectivity training. Open hand-drawn efforts such as CircuitNet [28] likewise rely on traditional image processing plus a classifier and a heuristic netlist generator.

### III. METHOD

#### A. PIPELINE OVERVIEW

Figure 1 summarizes the six-stage pipeline. The full pipeline proceeds through six stages: (1) component detection via oriented bounding box (OBB) YOLO model; (2) component occlusion (filling detected regions with local median color); (3) Sauvola adaptive binarization; (4) connected component labeling with morphological post-processing; (5) wire endpoint extraction via principal component analysis (PCA) on endpoint regions; and (6) wire joining via the endpoint-graph model with degree-budget completion. The output is a structural netlist mapping inferred component pins to electrical nodes. Component-pair evaluation does not certify those pin assignments or the exact net partition. SPICE export is illustrative: component values and device models require external specification, and simulation equivalence on real scans is not validated. Figure 2 shows two representative circuits at each stage.

### B. ENDPOINT-GRAPH JOIN MODEL

Given a set of wire segments  $\mathcal{W} = \{w_1, \dots, w_m\}$  with endpoints  $\text{ep}(w_i) = (a_i, b_i)$  and component pins  $\mathcal{P} = \{p_1, \dots, p_n\}$ , we construct an undirected graph  $G = (V, E)$  where  $V = \text{eps}(\mathcal{W}) \cup \mathcal{P}$ .

**Edge type 1 (wire body):** For each wire  $w_i$ , add edge  $(a_i, b_i)$  so both endpoints belong to the same net. The implementation accumulates connectivity with union-find; it does not solve a weighted graph optimization.

**Edge type 2 (endpoint–endpoint):** For endpoints  $e_i, e_j$  on different wires with  $\|e_i - e_j\| \leq \tau_{\text{join}}$ , add an edge. This proximity rule can connect fragments, corners and junctions, but can also merge unrelated nearby wires.

**Edge type 3 (endpoint–pin):** Each endpoint is connected to at most one selected pin by this edge rule. First apply the component-first assignment described below. Only if no component is assigned, search pins within Euclidean distance  $d \leq \tau_{\text{pin}}$  and select the pin with the smallest score. The score is  $d$  unless directional preference is enabled and  $d > 10^{-6}$  px; in that case it is  $d[1 - \alpha \max(0, \cos \theta)]$ , with fixed  $\alpha = 0.35$  and  $\theta$  measured from the outward wire direction to the endpoint-to-pin vector. Directional preference is enabled in the scale-relative base, but applies only to this fallback search. If an assigned component has no eligible pin, or the fallback finds none, no endpoint–pin edge is added. The score selects a fallback pin; it is not a weight optimized over the final graph.

**Edge type 4 (endpoint–wire-body):** For an endpoint  $e$  and a different wire segment  $w_j$ , add an edge if the point-to-segment distance is at most  $\tau_j$  and the normalized projection parameter lies in  $[0.15, 0.85]$ . This rule represents T-junction candidates away from the segment ends.

**Edge type 5 (pin–wire-body):** For a component pin  $p$  and wire segment  $w_j$ , add an edge if the point-to-segment distance is at most  $\tau_j$  and the normalized projection parameter lies in  $[0.05, 0.95]$ . This rule represents pin-to-rail candidates along a wire's body; geometric proximity alone does not certify an electrical connection.

The tolerances are manually configured scale-relative rules with fixed pixel clamps:  $\tau_{\text{pin}} = \min(60, \max(24, 0.62s))$ ,  $\tau_{\text{join}} = \min(28, \max(11, 0.30s))$ , and  $\tau_j = \min(20, \max(8, 0.20s))$  in pixels. Here  $s$  is the characteristic component-bounding-box diagonal scale; the implementation uses the middle sorted diagonal (the upper middle for an even count), falling back to a wire-length scale when component geometry is unavailable or degenerate. The multipliers and clamps are fixed, not learned. Because  $s$  is a single scalar per image, intra-image size variance is not modeled: a schematic mixing a large IC with small discrete symbols shares one tolerance per rule. Extreme-size performance was not evaluated by controlled image rescaling.

**Component-first pin assignment:** For edge type 3, we use a two-step assignment. First, compute the distance from each endpoint to each component's oriented bounding box (OBB): 0 inside the rectangle, otherwise the minimum distance to its edges. If four OBB vertices are unavailable, use axis-aligned bounding-box (AABB) distance instead. Assign the

endpoint to the nearest component whose distance is within  $\max(\tau_{\text{pin}}, 0.5 \cdot \text{diagonal})$ , where the diagonal is computed from that component's AABB. Second, within the assigned component, select the pin with the smallest Euclidean distance to the endpoint. Figure 3 contrasts the legacy radius join with our endpoint-graph model.

Nets are computed as connected components of  $G$ , projected onto pins. The result is a netlist mapping each pin to a net identifier.

### C. DEGREE-BUDGET COMPLETION

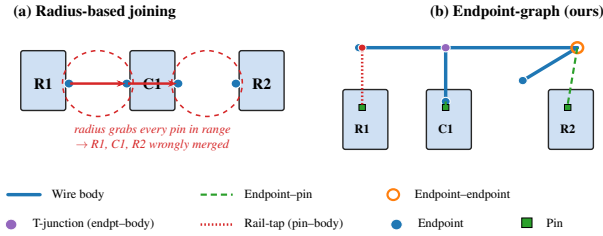
The endpoint-graph join may leave a pin in a net containing only pins from its own component. Completion treats such pins as candidates for recovery, excluding inert types such as ground, supply markers, text, antenna, probes, junctions and terminals. This condition can reflect a missing connection, but does not establish that the detector dropped a wire.

Degree-budget completion retains the base net assignments and constructs candidates from each floating pin  $p$  to pins  $q$  on other components and different current nets. Candidate distance must not exceed  $r = \rho \tau_c$ , where  $\tau_c = \min(60, \max(24, 0.62s))$  px and the deployed scale-relative completion uses  $\rho = 4$ . Thus reach uses the clamped pin scale, not the raw component diagonal; completion retains this scale dependence when base tolerances are ablated. For each candidate, a wire witness is sought by pairing the endpoints of a detected segment with  $p, q$  in either orientation. The witness cost is the smallest, over feasible segments and orientations, of the larger endpoint-to-pin distance, bounded by  $r$ . If no witness exists, the deployed relaxed-witness mode uses  $\|p - q\|$  as the candidate cost.

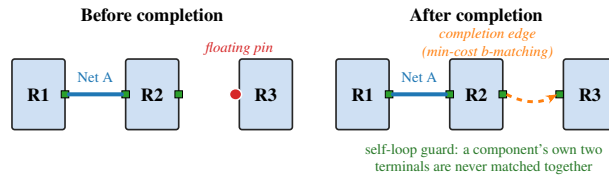
When candidates exist, the implementation builds a rectangular assignment matrix with one row per floating pin. Each target pin supplies up to three distinct slot columns, limited by its candidate demand, and each row has a private dummy column with cost  $1.5r$  for opting out. Unavailable assignments carry sentinel cost  $10^6$ . The Hungarian algorithm [24] (`scipy.optimize.linear_sum_assignment`) assigns every row to exactly one column, with at most one row per column. This mandatory assignment with dummy options avoids the empty-solution ambiguity of a nonnegative-cost objective with only at-most-one constraints; target slots implement the bounded-capacity matching [25].

Matches are then applied sequentially. Dummy or unavailable assignments add no edge. A proposed merge is also skipped if its endpoints are already in the same net or if their current nets share a component. This guard limits additional completion merges; it does not undo incorrect base unions or guarantee globally short-free output. Rejected matches are not reassigned, so the final guarded result is not claimed to minimize the assignment objective over all valid netlists.

Completion edges are “wireless”: they represent inferred connections without corresponding wire segments, analogous to manual pin merges in schematic editors. Figure 4 illustrates the completion step.



**FIGURE 3.** Joining approaches compared. (a) Radius-based join (legacy): each wire endpoint grabs all pins within a fixed threshold, causing over-merging. (b) Our endpoint-graph model: typed edges and component-first pin assignment generate connectivity candidates, with directional scoring confined to the fallback pin search. Incorrect base merges remain possible.



**FIGURE 4.** Degree-budget completion. Left: After base joining, some pins are “floating” (touching only their own component). Right: Completion reconnects floating pins via min-cost  $b$ -matching (orange dashed), with a guard rejecting additional merges between nets that already share a component; existing base shorts are not repaired.

## IV. RESULTS

### A. SYNTHETIC EVALUATION FRAMEWORK

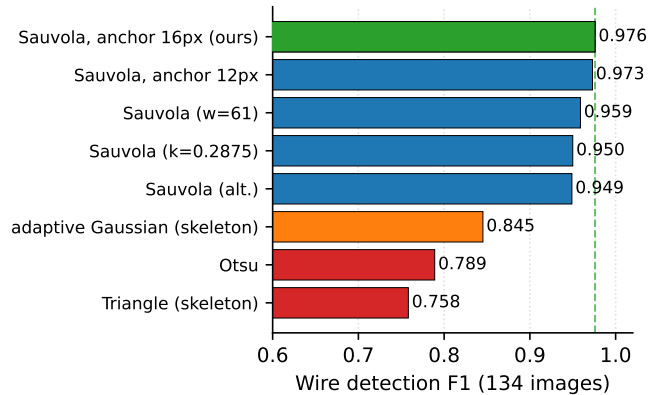
We author 15 synthetic circuits spanning 3–6 components and 2–6 nets, including parallel resistors, series dividers, Wheatstone bridges, RC/RL circuits, diode loops, and a 6-component ring (cyclic and bridge topologies being the main stress cases for join robustness). Each circuit has a known ground-truth netlist and SPICE deck with a test voltage source.

We inject five categories of error at four severity levels (L1–L4) on top of the clean (L0) baseline. These are Gaussian endpoint jitter with standard deviation  $\sigma$  (from 3 px at L1 to 14 px at L4), cut-short corruption in which each wire is shortened inward by a fixed length to model endpoints that stop short of the pin (from 4 px at L1 to 22 px at L4), full wire drops with probability  $p_{\text{drop}}$  (up to 0.20 at L4), wrong-pin snaps with probability  $p_{\text{wp}}$  (from 0.02 at L2 to 0.10 at L4), and anchor displacement, where with probability  $p_{\text{disp}}$  one endpoint is moved by up to  $a_{\text{max}}$  pixels (from 30 px at L1 to 80 px at L4).

We report join F1 as the component-connectivity F1 score, measuring which unordered pairs of components share a net in the predicted versus ground-truth netlist. This metric discards terminal identity: even a perfect component-pair score does not certify pin assignment, the exact net partition, absence of shorts, or simulation equivalence. We report simulation accuracy as the percentage of random seeds for which every source current in the predicted SPICE netlist matches the clean oracle within 5%.

### B. WIRE DETECTION BENCHMARK

We evaluate on 134 images from CGHD-1152 with ground-truth wire and component labels. Figure 5 and Table 1 sum-



**FIGURE 5.** Wire detection F1 across the tested thresholding methods and pipeline variants (134 images, annotated component boxes). The historical Sauvola configuration with a 16 px anchor filter and overlap deduplication at  $10^\circ/18$  px achieves F1 = 0.976; this is distinct from the production  $12^\circ/8$  px deduplication setting. The adaptive Gaussian and Triangle bars use the stored skeleton-extraction configurations (F1 = 0.845 and 0.758, respectively).

marize the top configurations across 36 tested variants.

The historical 16 px-anchor configuration achieves median F1 1.000, with 87% of benchmark images scoring F1 at least 0.90. Its aggregate F1 is 0.9755 at  $10^\circ/18$  px overlap deduplication; the  $12^\circ/8$  px production configuration gives 0.9726 with the same 16 px anchor distance. Among the tested alternatives, Otsu yields F1 0.789 and adaptive Gaussian 0.845. Occlusion is intended to remove component internals before wire extraction. Disabling that masking step while holding the rest of the extraction path fixed has a large measured downstream effect on the 31-image connectivity benchmark (Section IV-E): component-pair micro-F1 falls from 0.890 to 0.336. This is a mechanism ablation scored on downstream

**TABLE 1.** Wire detection performance on 134 images with annotated component boxes. The two Sauvola anchor-distance rows use historical 10°/18 px overlap deduplication; they are not the production 12°/8 px configuration.

| Configuration                         | F1           | Prec  | Rec   |
|---------------------------------------|--------------|-------|-------|
| Sauvola + anchor 16 px (ours)         | <b>0.976</b> | 0.973 | 0.978 |
| Sauvola + anchor 12 px                | 0.973        | 0.974 | 0.972 |
| Sauvola ( $w=61$ ) + anchor 12 px     | 0.959        | 0.944 | 0.974 |
| Sauvola ( $k=0.2875$ ) + anchor 14 px | 0.950        | 0.921 | 0.980 |
| Otsu threshold + CCL                  | 0.789        | 0.796 | 0.783 |

connectivity; it is not a wire-level F1 measurement for the extractor, whose human annotations are attached to the unmasked images.

### C. JOIN STRATEGY COMPARISON

Table 2 and Figure 6 compare joining strategies on the synthetic evaluation framework (15 circuits  $\times$  8 seeds, mean F1 over error levels L1–L4).

**TABLE 2.** Join Strategy Leaderboard (Synthetic Eval)

| Strategy                             | L0   | L1   | L2   | L3          | L4          |
|--------------------------------------|------|------|------|-------------|-------------|
| Scale-rel. graph + completion (ours) | 1.00 | 1.00 | 1.00 | <b>0.96</b> | <b>0.95</b> |
| Rescue graph + completion            | 1.00 | 1.00 | 1.00 | 0.95        | 0.94        |
| Rescue graph (base)                  | 1.00 | 1.00 | 1.00 | 0.94        | 0.90        |
| Scale-rel. graph (base)              | 1.00 | 1.00 | 1.00 | 0.94        | 0.85        |
| Radius union-find (legacy)           | 1.00 | 0.97 | 0.90 | 0.56        | 0.36        |
| Nearest-pin + bbox anchor            | 1.00 | 0.97 | 0.89 | 0.38        | 0.32        |

On the authored synthetic suite at L4, the scale-relative graph with completion scores F1 0.95, rescue graph with completion 0.94, rescue graph alone 0.90, scale-relative graph alone 0.85 and radius union-find 0.36 (Table 2). These results compare the reported configurations under the specified perturbations. They do not establish that completion repairs every dropped connection or prevents all shorts. The scale-relative completion configuration also has the highest component-pair score among the deterministic configurations reported on the real-image benchmark (Section IV-D). For readability, two intermediate synthetic variants whose scores round to rows shown are omitted: nearest-pin without anchoring, and the scale-relative base with end-extension but no dead-end rescue.

### D. REAL-IMAGE NET-LEVEL EVALUATION

The synthetic results above measure robustness under an authored error model. For the real-image evaluation, we constructed a human-verified net-level benchmark of 31 CGHD-1152 images using a purpose-built verification tool. Thirty-four images were annotated; three were excluded before scoring (one pending human verification, two failing the documented physical-sanity checks), leaving the scored  $N=31$ . Net-level proposals were bootstrapped by a conservative endpoint-graph join over the dataset’s annotated wires and

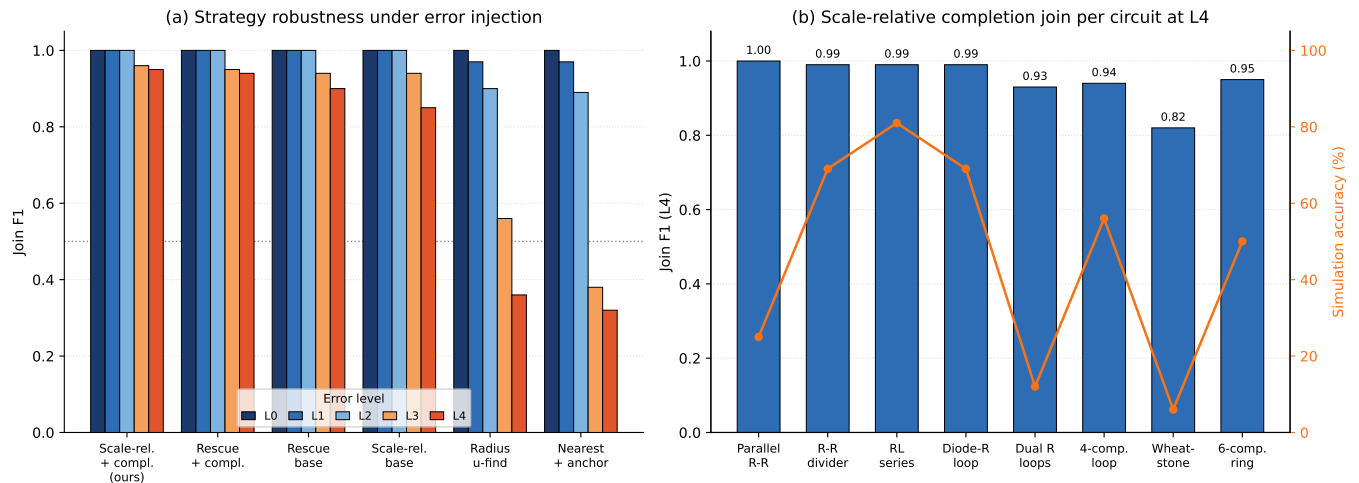
then corrected by one human annotator. Human correction and evaluation on independently authored synthetic ground truth mitigate concern about method-dependent labels, but do not rule out residual bootstrap or annotator bias in the real benchmark. Both the geometric join and the VLM diagnostic receive annotated component boxes. Connectivity is scored as component-pair F1 restricted to the evaluated electrical-component subset, discarding terminal identities as in the synthetic connectivity metric.

Table 5 reports the evaluated join configurations on the verified set, run on the pipeline’s own detected wires. We report micro-averaged (pair-pooled) component-pair F1 as the primary metric, so that a dense 14-component board contributes in proportion to its connections rather than counting the same as a 3-component one; the macro (mean per-image) F1 is listed alongside for reference. The best configuration, the scale-relative completion join (degree-budget completion applied to a high-precision scale-relative endpoint-graph base), achieves micro-F1 = 0.890 ( $P = 0.92$ ,  $R = 0.86$ ; 95% bootstrap CI [0.855, 0.924]), improving on the prior default that applies the same completion to the rescue base (0.829) and the radius union-find join (0.667) (Figure 7). The scale-relative completion configuration omits the rescue base’s 12 px end-extension and dead-end rescue and has higher reported precision. This comparison does not isolate the causal contribution of those individual changes. A bounded within-benchmark reach sweep over  $\rho \in [3, 5]$ , with  $r = \rho\tau_c$ , yields macro-F1 0.895–0.903. This is sensitivity evidence for the tested settings, not a controlled image-rescaling or mixed-size robustness experiment. Confidence intervals are wide at  $N = 31$  and the per-strategy intervals overlap, but the ranking is consistent and is corroborated on the independently authored synthetic suite (Table 2).

**TABLE 3.** Leave-one-out ablation of endpoint-graph rules at the base-graph level and with completion, using annotated component boxes and detected wires on the 31-image benchmark. All full-pipeline rows have the same aggregate TP/FP/FN (418/37/66). Fixed-pixel base tolerances give micro-F1 0.820 versus 0.816 for the scale-relative base; completion retains its scale dependence. Aggregate ties do not establish identical recovered connections or a causal explanation for the ties.

| Configuration                                     | Base graph only |            |
|---------------------------------------------------|-----------------|------------|
|                                                   | micro-F1        | TP/FP/FN   |
| Baseline (all edge types, scale-relative)         | 0.816           | 352/27/132 |
| T-junctions disabled                              | 0.816           | 352/27/132 |
| Rail taps disabled                                | 0.816           | 352/27/132 |
| T-junctions + rail taps disabled                  | 0.816           | 352/27/132 |
| Directional preference disabled                   | 0.816           | 352/27/132 |
| Fixed-pixel base tolerances; completion unchanged | 0.820           | 356/28/128 |

Table 3 reports bounded interventions on the base graph and the resulting pipeline with completion. Disabling T-junctions, rail taps or directional preference leaves the displayed base and full aggregate counts unchanged. Fixed-pixel base tolerances increase base micro-F1 from 0.816 to 0.820; the full-pipeline counts again tie at 418/37/66. Equal aggregate counts do not establish identical recovered connections, explain which mechanisms were active, or prove that completion masks differences. The baseline comparison shows an



**FIGURE 6.** (a) Join strategy robustness under error injection (15 circuits  $\times$  8 seeds): bars give mean connectivity F1 at each severity level (L0–L4, dark to light to red). The completion-based strategies (left) stay near 1.0, while the radius union-find and nearest-pin baselines (right) collapse at L3–L4. (b) The scale-relative completion join per circuit at L4 (16 seeds): bars are connectivity F1, the orange line is simulation accuracy. High precision does not exclude false merges or certify pin correctness; the Wheatstone bridge has the lowest reported connectivity F1 and simulation accuracy (Table 7).

increase from base micro-F1 0.816 to 0.890 with completion, but these ablations do not establish that other mechanisms are generally unnecessary. Full occlusion removal and the completion safeguards are tested separately (Table 4, Section IV-E). Removal of scale dependence throughout completion, and extreme-size or mixed-size input conditions, were not tested. The ablation evidence therefore still does not cover every proposed module or every input condition.

#### E. OCCLUSION REMOVAL AND COMPLETION-SAFEGUARD ABLATIONS

Table 4 reports the mechanism interventions that Table 3 does not cover, evaluated on the same 31-image benchmark with annotated component boxes and detected wires.

**TABLE 4.** Mechanism ablations on the 31-image benchmark. The occlusion arm removes only the component-masking call from the extraction path; ROI cropping, extractor configuration and coordinate restoration are held fixed. Guard and witness arms act on `scale_completion`. “Images changed” counts images whose predicted component pairs differ from the reference arm.

| Arm                      | TP/FP/FN   | micro-F1     | macro-F1 | Images changed |
|--------------------------|------------|--------------|----------|----------------|
| Occlusion on (reference) | 418/37/66  | <b>0.890</b> | 0.901    | —              |
| Occlusion off            | 103/26/381 | 0.336        | 0.294    | 31/31          |
| Guard on (reference)     | 418/37/66  | <b>0.890</b> | 0.901    | —              |
| Guard off                | 418/37/66  | 0.890        | 0.901    | 0/31           |
| Witness required         | 413/37/71  | 0.884        | 0.907    | 6/31           |

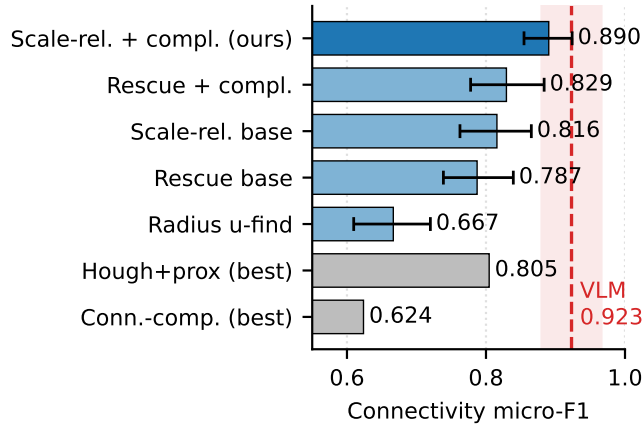
Removing component occlusion leaves component coordinates available for region-of-interest selection and anchor filtering but exposes the component pixels to the extractor, which then returns 466 wire segments instead of 666. Component-pair micro-F1 falls from 0.890 to 0.336 (precision 0.798, recall 0.213) and predicted pairs change on all 31 images. This establishes a large measured downstream contribution for the occlusion step on this benchmark. It is a mechanism ablation, not a wire-level extractor score: no

validated one-to-one wire-matching scorer was run, and the human wire annotations are attached to the unmasked images, so wire-level F1 is not reported. The occlusion-off arm is not a proposed configuration and is not a replacement estimate for the pipeline’s benchmark performance.

For the completion safeguards, the shared-component (self-loop) guard fires on 18 of 31 images and rejects 38 candidate merges. Because the primary metric projects each net to distinct component pairs, disabling the guard leaves the scored pair sets unchanged: the guard’s protection against same-device-pin shorts is invisible to this metric and therefore unquantified here. Requiring a wire witness instead of the relaxed distance fallback changes predictions on 6 of 31 images (TP/FP/FN 413/37/71; micro-F1 0.884, macro-F1 0.907). These arms establish activation and paired effects; they do not rank the safeguards by safety.

**TABLE 5.** Join strategy and baseline comparison on human-verified net-level ground truth (31 real images, annotated component boxes; detected wires except for the stated perfect-wire reference). Primary metric is micro-averaged (pair-pooled) component-pair F1; macro (mean per-image) F1 is shown for reference.

| Method                                                              | F1           | Prec  | Rec   | F1 <sub>macro</sub> |
|---------------------------------------------------------------------|--------------|-------|-------|---------------------|
| <i>Deterministic joins (ours and ablations)</i>                     |              |       |       |                     |
| Scale-rel. graph + completion (ours)                                | <b>0.890</b> | 0.919 | 0.864 | <b>0.901</b>        |
| Rescue graph + completion (prior)                                   | 0.829        | 0.789 | 0.874 | 0.853               |
| Scale-rel. graph (base)                                             | 0.816        | 0.929 | 0.727 | 0.838               |
| Rescue graph (base)                                                 | 0.787        | 0.811 | 0.764 | 0.813               |
| Radius union-find (legacy)                                          | 0.667        | 0.698 | 0.638 | 0.674               |
| <i>Classical deterministic baselines (identical detected wires)</i> |              |       |       |                     |
| Hough + proximity (best)                                            | 0.805        | 0.750 | 0.868 | 0.847               |
| Connected-components (best)                                         | 0.624        | 0.965 | 0.461 | 0.611               |
| <i>Conditional VLM diagnostic / annotated-wire reference</i>        |              |       |       |                     |
| VLM (Claude Opus 4.8)                                               | 0.923        | 0.970 | 0.880 | 0.949               |
| Scale-rel. + completion, perfect wires                              | 0.890        | 0.913 | 0.868 | 0.916               |

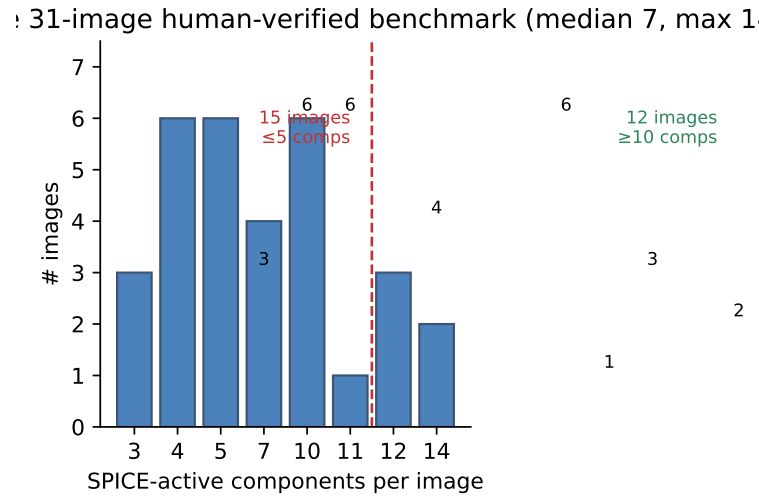


**FIGURE 7.** Micro-averaged connectivity F1 on the 31-image human-verified net-level benchmark (component-pair metric). The scale-relative completion join has the highest score among the evaluated deterministic configurations and classical baselines on these detected wires. The dashed line marks the VLM diagnostic (Claude Opus 4.8). Both methods receive annotated component boxes; the VLM-minus-join difference is  $+0.033$  (95% bootstrap CI  $[-0.009, +0.078]$ ). The interval does not establish equivalence, cost superiority or a global validity guarantee. Error bars are 95% bootstrap CIs.

This real-image ranking resolves an open question from prior work on this pipeline: against bootstrapped pilot labels the degree-budget completion had *not* consistently beaten its base join; against human-verified ground truth it clearly does (micro-F1 0.829 vs 0.787 for the rescue base alone), and the scale-relative variant is better still. The same ordering holds on the authored synthetic suite (Table 2), where the scale-relative completion join also ranks first, providing corroborating evidence on independently authored data without ruling out residual bias in the real benchmark labels.

Several image-to-netlist systems erase component regions and infer nets from the remaining wire pixels using connected components or contours, including SINA [3] and Peker *et al.* [5]. We evaluate a connected-component baseline on our detected wire segments to control the wire input. Its best tested configuration reaches component-pair micro-F1 0.624, versus 0.890 for the endpoint-graph join (Table 5). Hough line fitting [20] followed by proximity linking reaches 0.805 at its best tested linking tolerance on the same detected wires. These are the best configurations in the reported sweeps, not upper bounds on all implementations of the classical methods. Their ordering is evidence for these configurations and this benchmark, not a direct performance comparison with the cited complete systems.

We additionally attempted to run five public image-to-netlist systems on the hand-drawn benchmark and encountered the following implementation, model-vocabulary and input-domain limitations. The classical printed-schematic interpreter of Kelly and Cole [2] installs and runs, but on hand-drawn scans it over-segments wires into spurious components and recovers almost no connectivity, a domain-transfer failure. Netlistify [27] reproduces on its own printed AMS test data, which we ran end-to-end with the released model, but



**FIGURE 8.** Complexity of the 31-image human-verified benchmark, counting components in the evaluated electrical subset: 15 images have  $\leq 5$  and 12 have  $\geq 10$  (median 7, range 3–14). Component-count breadth does not validate dense buses or multilayer crossings; performance on these cases remains untested.

its active model lacks the diode, inductor, and IC classes our circuits contain and its connectivity model is trained on printed thin-line appearance, limiting its applicability to the device population in this benchmark. Masala-CHAI [4] delegates connectivity to a proprietary LLM behind a paid API, equivalent in spirit to our VLM experiment in Section IV-H, rather than a deterministic algorithm. SINA’s [3] anonymized code archive was unextractable and CircuitNet [28] ships only Colab notebooks without released weights. These observations concern the versions and configurations examined; they do not establish that no prior system can recover hand-drawn connectivity. The CCL and Hough experiments above provide controlled comparisons on the same detected wires.

With annotated component boxes held fixed, replacing detected wires by the dataset’s annotated wires changes component-pair micro-F1 from 0.8903 to 0.8898, while macro-F1 increases from 0.901 to 0.916 (Table 5). The micro scores round to the same displayed value, but are not identical. This comparison concerns wire input conditional on component annotations; it does not establish an autonomous detection bottleneck or exclude compensating changes in recovered connections.

#### F. DESCRIPTIVE PER-DRAFTER RESULTS

Table 6 reports the scale-relative completion join grouped by CGHD drafter. The published CGHD-1152 dataset file index (version 14) places each of the 31 benchmark image filenames under exactly one `drafter_N/images/` path, matching the drafter assignments used here. Thus all 31 assignments are direct matches to the corpus’s drafter labels rather than inferences from circuit numbering; the version 13 image-path index agrees. Four groups contain at least four images, with micro-F1 ranging from 0.854 to 0.987. These groups describe

within-corpus variation, not evidence of unseen-drafter transfer. No held-out-drafter experiment was performed; the small groups do not support strong statistical separation.

**TABLE 6. Descriptive join micro-F1 (scale-relative completion) by CGHD drafter. Groups with at least four images; overall includes all benchmark images. Drafter identities are direct filename matches to the published CGHD-1152 file index (31/31).**

| Drafter    | <i>n</i> images | micro-F1 | macro-F1 | TP/FP/FN  |
|------------|-----------------|----------|----------|-----------|
| drafter_1  | 7               | 0.933    | 0.907    | 56/6/2    |
| drafter_2  | 5               | 0.854    | 0.874    | 85/14/15  |
| drafter_10 | 5               | 0.987    | 0.995    | 37/1/0    |
| drafter_12 | 4               | 0.913    | 0.938    | 21/0/4    |
| Overall    | 31              | 0.890    | 0.901    | 418/37/66 |

## G. PER-CIRCUIT ANALYSIS

Table 7 shows the scale-relative completion join's performance across eight representative circuits from the 15-circuit synthetic corpus at L4 (extreme error injection, 16 seeds).

**TABLE 7. Scale-Relative Completion Join Performance per Circuit at L4 (16 seeds)**

| Circuit           | Comps | F1   | Prec | Sim. acc. |
|-------------------|-------|------|------|-----------|
| Parallel R-R      | 3     | 1.00 | 1.00 | 25%       |
| R-R divider       | 3     | 0.99 | 1.00 | 69%       |
| RL series         | 3     | 0.99 | 1.00 | 81%       |
| Diode-R loop      | 3     | 0.99 | 1.00 | 69%       |
| Dual R loops      | 4     | 0.93 | 0.90 | 12%       |
| 4-component loop  | 4     | 0.94 | 0.91 | 56%       |
| Wheatstone bridge | 6     | 0.82 | 0.85 | 6%        |
| 6-component ring  | 6     | 0.95 | 0.92 | 50%       |

Precision is at least 0.85 across the reported synthetic circuits at L4. This does not exclude false merges or establish that all errors are fragmentation. The Wheatstone bridge has the lowest reported F1 (0.82) and simulation accuracy of 6%, illustrating that component-pair connectivity and the separately evaluated synthetic simulation criterion measure different properties.

## H. VLM CONNECTIVITY EXPERIMENT

We evaluate Claude Opus 4.8 as an oracle-component-box connectivity diagnostic. Both methods receive the same annotated component boxes; the geometric method uses detected wires. The VLM is given the scanned image with component boxes labelled by index and asked to return JSON sets of component terminals sharing an electrical node. Each image uses a separate model call without shared conversational context. We score the resulting unordered component pairs against the same human-verified benchmark. This is a conditional connectivity comparison, not an autonomous end-to-end comparison.

On the 31 real images, the VLM reaches micro-F1 0.923 (precision 0.970, recall 0.880; macro-F1 0.949), versus micro-F1 0.890 for the geometric join, and is exact under the component-pair metric on 21/31 images. The paired

VLM-minus-join difference is +0.033, with 95% bootstrap CI  $[-0.009, +0.078]$ . Inclusion of zero does not establish equivalence. These results do not establish comparative cost superiority or a global guarantee against incorrect merges for either method. Autonomous performance requires separately auditable evaluation; the conditional scores do not certify pin-level or simulation correctness.

## V. DISCUSSION

The endpoint graph represents wire endpoints and component pins with separate rules for endpoint proximity, component assignment and mid-span contacts. Component-first assignment selects a nearby component by OBB distance and then a pin by Euclidean distance. These geometric choices provide inspectable connectivity candidates, but do not establish that a nearby wire is electrically connected or that a selected pin has the correct device-specific meaning.

Degree-budget completion proposes connections for pins left in single-component nets. Each floating-pin assignment selects at most one recovery target, and the shared-component guard can reject additional merges. Neither mechanism repairs shorts already present in the base graph. Wire-detection F1 does not measure an endpoint-drop rate or establish the causal contribution of missing endpoints to simulation failures.

The human-verified net-level evaluation, while novel for hand-drawn circuits, currently covers 31 images; the 95% bootstrap confidence intervals on the per-strategy F1 are correspondingly wide ( $\pm 0.03$ – $0.05$ ) and overlap between adjacent strategies, so while the top ranking is consistent and corroborated by the independently authored synthetic suite, broader annotation is needed to separate the closely ranked configurations. The net-level labels were bootstrapped by a conservative endpoint-graph join over perfect wires and then human-corrected; we mitigate the residual concern that this favors graph-based methods by validating on synthetic ground truth that is authored from scratch, where the same ranking holds. Annotation was performed by a single human annotator using a purpose-built verification tool, and no per-image timing log was retained, so a per-image annotation-cost figure cannot be reported. A proposed expansion protocol would stratify sampling by component count, drafter, crossing/bus structure and capture conditions, use independent annotation followed by adjudication, and record per-image effort and provenance logs. This is a proposed expansion protocol, not completed annotation. The evaluation draws from a single hand-drawn corpus (CGHD-1152); generalization across other drawing styles, capture conditions, and printed schematics remains to be validated.

Table 8 distinguishes the evaluated connectivity scope from generic pin guesses and illustrative export. Listing a device group does not establish separately detected model classes: the detector vocabulary merges multiple original classes. Pin construction depends on the component type supplied to the netlist stage; a template for a named complex device does not recover that identity from a merged detector label. Generic

pin geometry is not evidence of validated device-specific pin relationships.

Component detection has released-checkpoint mAP@0.5 of 89.0% (the checkpoint embeds 0.88977 at its best-fitness epoch; the previously reported 88.5% belongs to the final training epoch), with crossover recall of 70.7%. Geometric edge types 2, 4 and 5 do not consult crossover class labels when joining nearby wire fragments or mid-span contacts, creating a possible false-merge pathway. An audit using annotated component boxes found 13 crossover instances across eight benchmark images. In C242\_D1\_P1, suppressing two crossing-adjacent endpoint–endpoint edges changed component-pair TP/FP/FN from 27/4/0 to 19/0/8. Removing these edges also removes true connections; this is not a demonstrated mitigation gain. To test the detector half of this pathway directly, we ran the trained detector over the 31 benchmark images and compared its crossover outputs with the 13 annotated instances. Every annotated crossover was localized and assigned to the crossover class at a confidence of at least 0.25 (confidence 0.28–0.97, IoU 0.43–0.91); at a 0.50 threshold, 12 of 13 are retained and one falls below threshold. This does not contradict the reported 70.7% class recall, which is measured on the detector’s own validation data at its deployment threshold, and training-split membership for these benchmark images is unknown, so these are not held-out estimates. As a paired counterfactual, deleting the annotated crossover boxes before joining changes the recovered predictions on 6 of 31 images and changes component-pair micro-F1 by +0.010, with identical net partitions on 23 of 31 images. These two conditional checks therefore give no evidence that crossover misclassification is the driver of the false pairs observed here. Both are annotated-box, within-benchmark diagnostics: they are not a prevalence estimate, not an end-to-end error rate, and not a validation of crossover-aware suppression, which remains unvalidated.

The VLM comparison uses a single model (Claude Opus 4.8) on 31 images; evaluating additional models and prompting strategies is future work. The component detector was trained on CGHD-1152 after excluding drafter\_0, whose drawing style differed from the retained drafters; performance on that excluded style is not established here. The synthetic error model is an authored approximation rather than a calibrated fit to detector errors. Calibration to measured errors is future work.

## VI. CONCLUSION

We present structural circuit-netlist recovery using an occlusion-first wire extractor and endpoint-graph joining with degree-budget completion. On the 31-image within-corpus benchmark with annotated component boxes, component-pair micro-F1 is 0.8903; perfect-wire input gives 0.8898. These scores round alike but do not establish an autonomous detection bottleneck. The component-pair metric does not certify pin identities, exact net partitions, absence of shorts, or simulation equivalence. Synthetic L4 connectivity F1 is 0.95 versus 0.36 for the radius baseline. The VLM compar-

ison is conditional on annotated component boxes and does not establish equivalence or cost superiority. Export remains illustrative: values and device models require external specification, and real-scan simulation equivalence is unvalidated. Broader styles, dense buses, and mixed-scale schematics require future evaluation.

## ACKNOWLEDGMENT

The authors used Claude Opus 4.8 (Anthropic) to assist with grammar and language editing of the manuscript and as a programming assistant during development of the pipeline and evaluation code. This assistance is distinct from the VLM connectivity experiment in Section IV-H, where the same model is an evaluation subject. All methods, experiments, and conclusions are the authors’ own, and the authors reviewed and take full responsibility for the entire content of this article.

## DATA AND CODE AVAILABILITY

The pipeline, evaluation scripts, synthetic-circuit generator, the 31-image human-verified net-level ground truth (ground\_truth/real\_nets\_verified.json), the net-level verification tool used to produce it, the join-strategy and connected-component baseline benchmarks, and the Claude-VLM connectivity harness are available at <https://github.com/boscochanam/circuit-digitization>, including the per-image to CGHD drafter mapping used for the per-drafter analysis (Table 6). The component-detection model is available at <https://huggingface.co/boscochanam/circuit-component-detector>, trained on the CGHD-1152 dataset [1].

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**TABLE 8.** Connectivity evaluation, generic pin construction, and illustrative export are distinct capabilities.

| Device group                                                     | Evaluated scope                                                                        | Pin construction                                                                                                     | Export limitation                                                                                                                                     |
|------------------------------------------------------------------|----------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|
| R/C/L/D/Q, voltage sources and ICs                               | Component-pair connectivity on the benchmark; no universal validation of pin templates | Geometric templates; endpoint clustering for eligible discrete/source types, generic layouts for ICs                 | Illustrative defaults/substitutions; values and device models require external specification. ICs are skipped because no export model is implemented. |
| Switches                                                         | Absent from the evaluated electrical subset; switch-specific connectivity unvalidated  | Two pins from the shorter OBB-edge midpoints; long-axis AABB fallback                                                | Fixed $0.001\ \Omega$ resistor if pins 0 and 1 map to distinct nodes; otherwise no switch line. No switch state or control model is recovered.        |
| Transformers, thyristors, optocouplers and other complex devices | Device-specific connectivity unvalidated                                               | Named transformer type receives a two-pin geometric guess; optocoupler template and unknown-type fallback also exist | These complex types have no implemented export model and are skipped; geometric guesses do not establish winding, control or isolation semantics.     |
| Metric boundary (all evaluated groups)                           | Component pairs discard terminal identity                                              | F1 does not certify pin assignment or the exact net partition                                                        | F1 does not certify absence of shorts or simulation equivalence.                                                                                      |

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